AYMANE AIT DADS

AI Research Intern - LLM Fine-Tuning & Reinforcement Learning | Transformer Architectures · Supervised Fine-Tuning · Deep Learning Research

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PROFESSIONAL SUMMARY

Achieved top 10% public leaderboard (0.80/1.0) in the NVIDIA Nemotron Reasoning Kaggle competition ($106K+ prize pool) by fine-tuning a 30B Mamba-Transformer MoE via LoRA, conducting 20+ single-variable ablation studies to identify optimal training configurations for chain-of-thought reasoning. Ranked 1st in EURECOM class (0.791 AUC) on an AI-generated image detection benchmark and submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark, demonstrating research output aligned with top-tier conference standards. Brings deep expertise in transformer architectures, supervised fine-tuning, PyTorch, and optimization methods for large language models, with applied experience in NLP pipelines, computer vision, and CUDA-level GPU debugging on Blackwell hardware. Available for a 6-month internship in 2026 in the Asia region; no visa restrictions anticipated.

CORE COMPETENCIES

Reinforcement Learning | LLM Agent Research | Supervised Fine-Tuning | Transformer Architectures | Deep Learning Optimization | Natural Language Processing | Ablation Studies | PyTorch & Python Proficiency

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Engineered a supervised Random Forest classification pipeline on **100K+ fiber-optic network records**, mapping latency, jitter, and upload speed features to predictive quality labels across thousands of nodes.
- Developed an unsupervised K-Means clustering pipeline segmenting customer performance profiles into **5 commercial service tiers**, with output adopted directly by the network optimization team for infrastructure decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and reproducible Python-based EDA scripts built with Pandas and Scikit-learn.
- Delivered stakeholder presentations translating **machine learning model outputs** into actionable maintenance recommendations, using Power BI dashboards to communicate findings to non-technical audiences.

PROJECTS

NVIDIA Nemotron Reasoning Challenge - Kaggle Competition
Kaggle Competition · In Progress (June 2026) · Top 10% Public Leaderboard · $106K+ Prize Pool

- Fine-tuned a **30B Mamba-Transformer MoE (3.5B active via MoE routing)** using LoRA (r=32, ~888M trainable parameters) on 7,828 verified chain-of-thought examples, achieving a public leaderboard score of 0.80/1.0 (top 10%) in an active $106K+ prize-pool competition.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict sequence - including monkey-patched CUDA allocator warmup, stubbed Mamba/Cutlass imports, and a pure PyTorch RMSNorm fallback - and conducted 20+ single-variable ablation studies confirming completion-only loss (+0.01 AUC) and that sequence packing caused a 0.16-point score collapse.

PyTorch · Unsloth · TRL/PEFT · Hugging Face Transformers · vLLM · Triton · Mamba

AI-Generated Image Detection - NTIRE 2026 @ CVPR & EURECOM ImSecu
EURECOM Course + International Benchmark · 2025-2026 · 1st in Class Leaderboard

- Fine-tuned a **CLIP ViT-L/14 (428M params)** backbone with a custom MLP head using dual learning rates (2e-6 CLIP, 5e-4 head) and 3-model ensemble with 10-view TTA, achieving 0.791 AUC and ranking 1st on the EURECOM private class Kaggle leaderboard.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; ablation over unfreezing 1/4/6/8 transformer blocks showed 1-epoch training (0.793 AUC) outperformed 4-epoch (0.767 AUC), confirming out-of-distribution generalization as the dominant factor.

PyTorch · OpenCLIP · Hugging Face Transformers · FP16 Mixed Precision · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM — *Sep 2023 - Present*

Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Statistics, Image Security, Cloud Computing, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya — *2021 - 2023*

SKILLS

LLM Fine-Tuning & NLP: LoRA / QLoRA / PEFT | SFTTrainer (Unsloth) | Completion-Only Loss | Mamba-Transformer / MoE | vLLM | Chain-of-Thought | Hugging Face Transformers

Deep Learning & Computer Vision: PyTorch | CLIP / ViT | CNN / DenseNet | FP16/BF16 Mixed Precision | TRL | TensorFlow | Scikit-learn

MLOps, Data Science & Infrastructure: CUDA Debugging | Triton | Ablation Studies | Pandas / NumPy | Git / Linux / Bash | Power BI | Docker (basic)